

Lesson Notes

MODULE 5 | LESSON 3

LIQUIDITY AND THE CREDIT MARKET

Reading Time	75 minutes
Prior Knowledge	Bond, Issuance, Probability of Default (PD)
Keywords	Credit market, Credit rating, Credit Spread (CS), Credit Default Swap (CDS), Liquidity, Recovery Rate (CS), Probability of Default (PD), Bid-ask spread, Fungibility

In the last lesson, we applied the 5 Cs of credit analysis to see which of two MBS offerings had better credit qualities. We discussed how a mortgage exhibits nonlinearity, similar to other derivatives we have seen (call and put options). We witnessed how model risk contributed to the Great Financial Crisis. We discussed the impact of moral hazards (such as information asymmetry and conflict of interest) in the mortgage and bond markets and how this led to the need for new regulations such as Dodd-Frank in the U.S.

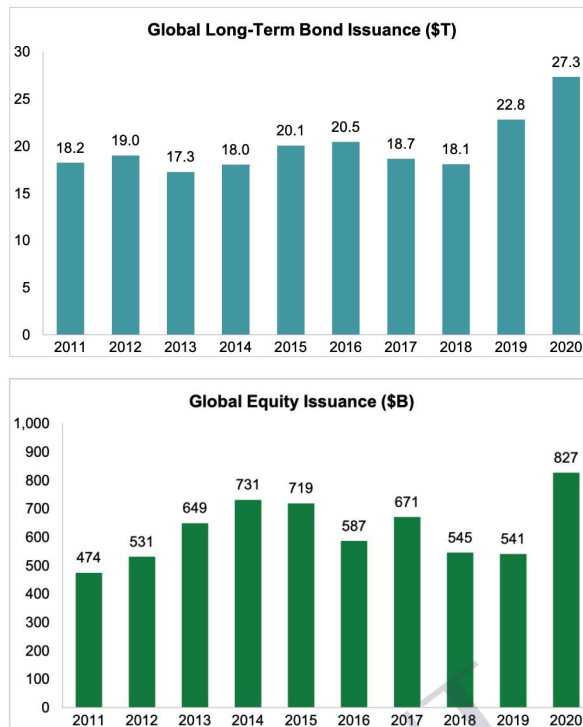
In this lesson, we quantify the size of the credit market, both overall as well as by country. We learn how to calculate a credit spread based on just a few inputs and how to "back out," or calculate the implied probability of default from a bond's credit spread. We discover the CDS (credit default swap) market as a way to trade the credit risk of a bond or bond issuer. We also learn the relationships between rating and probability of default and between rating and credit spread. We also discuss one of the big problems with certain risk measures and with credit ratings. Finally, we discuss some characteristics of liquidity and how we might estimate it.

1. The Credit Market and Its Components

The credit market includes securities that involve some sort of borrowing or debt. For this reason, the "debt market" is synonymous with the "credit market." Since the debt market comprises mostly bonds, the credit or debt market may also be known as the bond market.

The bond market is vast, even compared to the global equity market. Compare the annual issuance of debt and equity in the following two graphs and be sure to note the units (\$T for bonds vs. \$B for equity). How many times larger was bond issuance in 2020 compared to equity issuance?

Figure 1: Bond and Equity Issuance



Sources: Bloomberg, Refinitiv, Dealogic
 Note: Long-term is defined as a security with maturity of 13 months or longer at issuance; global equity issuance includes non-convertible IPOs and follow-on equity deals, excludes preferred shares, rights issued, closed-end funds, business development companies, and special purpose acquisition companies.

Source: Kolchin, Katie et al. 2021 *Capital Markets Fact Book*. SIFMA, July 2021, p. 38. <https://www.sifma.org/wp-content/uploads/2021/07/CM-Fact-Book-2021-SIFMA.pdf>.

Compare bond issuance in 2020 to 2018. What is this ratio? It appears to be exactly 1.5x. This suggests that the bond market was very healthy in 2020, despite the beginning of the global COVID-19 pandemic. Interestingly, the equity market also appears to have been rather healthy in terms of global issuance. (How the bond and equity markets *performed* in terms of the profit earned by investors in those markets is a completely separate question, which we do not address in this lesson.)

The following tables provide a different but related view: the value of the total outstanding bonds. Which countries had the five largest values of bonds outstanding in 2020? Let's consider the European Union (EU), a single country for this purpose.

Figure 2: Outstanding Bond Value

	\$ Billions										
	Australia	Canada	China	EU27	Hong Kong	Japan	Singapore	United Kingdom	United States	Other Developed Markets	Emerging Markets
2006	957.3	1,327.4	1,198.3	15,894.2	99.1	9,090.3	140.3	3,989.8	23,881.7	527.1	633.9
2007	1,251.4	1,542.9	1,704.5	19,312.8	104.3	9,852.1	166.3	4,708.8	26,145.9	602.2	808.9
2008	1,141.9	1,369.7	2,228.0	20,256.7	102.7	12,329.9	167.6	4,099.6	28,033.8	604.0	749.0
2009	1,523.7	1,671.6	2,577.9	24,101.1	151.3	12,268.5	185.0	5,280.9	29,132.6	775.9	840.7
2010	1,742.9	1,866.6	3,065.3	23,324.9	184.0	14,604.1	218.0	5,210.2	30,208.7	833.7	991.9
2011	1,872.6	1,963.4	3,528.4	23,463.5	211.7	15,009.1	240.9	5,587.2	31,021.5	839.0	1,001.7
2012	2,037.5	2,173.5	4,294.1	24,307.6	250.5	14,254.1	294.8	5,828.8	32,300.4	950.8	1,140.0
2013	1,879.5	2,241.9	4,960.8	24,999.2	303.9	11,909.2	331.4	5,851.0	33,284.5	960.2	1,100.9
2014	1,916.9	2,206.1	5,835.4	22,074.9	343.5	10,558.0	349.3	6,169.5	34,302.9	888.8	1,150.4
2015	1,819.1	2,767.5	7,752.5	19,871.1	372.4	10,638.7	322.3	5,891.5	35,317.4	859.9	1,199.2
2016	1,823.2	2,943.5	9,408.8	19,379.9	404.9	11,249.5	326.4	5,439.2	36,716.1	935.1	1,272.1
2017	2,003.9	3,267.4	11,757.1	22,129.5	473.7	11,932.1	412.2	6,078.4	37,903.4	1,033.3	1,497.8
2018	1,899.9	3,149.3	12,906.9	21,437.8	511.7	12,477.0	441.1	5,792.2	39,456.1	1,003.6	1,501.9
2019	1,957.4	3,369.5	14,725.9	21,788.2	531.9	12,828.8	492.9	6,325.2	41,226.0	1,117.9	1,601.4
2020	2,498.4	3,910.3	18,555.6	25,226.0	556.1	14,678.3	542.8	7,172.4	47,237.0	1,334.8	1,757.0
Average	1,755.0	2,384.7	6,966.6	21,837.8	306.8	12,296.6	308.8	5,568.2	33,744.5	884.4	1,149.8
Y/Y % Change	27.5%	16.1%	26.0%	15.8%	4.6%	14.4%	10.1%	13.4%	14.6%	19.4%	16.5%
5-Year CAGR	6.6%	7.2%	19.1%	4.9%	8.3%	6.6%	11.0%	4.0%	6.0%	9.2%	7.3%
10-Year CAGR	3.7%	7.7%	19.7%	0.8%	11.7%	0.1%	9.5%	3.2%	4.6%	4.8%	5.9%

Source: Kolchin, Katie et al., 2021 *Capital Markets Fact Book*, SIFMA, July 2021, p. 44. <https://www.sifma.org/wp-content/uploads/2021/07/CM-Fact-Book-2021-SIFMA.pdf>.

At \$47 trillion (about 38% of the \$123 trillion global total), the United States' outstanding bond value is almost twice as much as the EU's (about 20% of the global total). These two are followed by China (15%), Japan (12%), and the United Kingdom (6%). The rest of the bond markets by country are rather small compared to these five large ones.

The required reading "[Bonds](#)" defines liquidity risk as "the risk that investors won't find a market for the bond, potentially preventing them from buying or selling when they want" (U.S. Securities and Exchange Commission). (We discuss liquidity in more detail later in this lesson.) If you were an investor looking for a bond market with low liquidity risk, you would probably start looking at one of the five large bond markets. You still need to take into account the liquidity of the specific bond within these markets, but you are more likely to find a liquid bond in these markets than in some of the other much smaller bond markets. If you had to base a decision on where to invest to

minimize liquidity risk based only on the information in these two charts, would you start looking for a bond investment in Hong Kong or in Australia?

On the other hand, you may be willing to take on some liquidity risk, as long as you are appropriately compensated for that risk. Depending on your preferences as an investor, there is probably *some* level of bond yield that would induce you to invest in a less liquid bond or bond market. So the smaller bond markets are not necessarily less attractive, so long as you are compensated for the risk you take. Risk and return generally go hand in hand. This is one meaning of the economic adage, "There is no such thing as a free lunch." So if you "had appetite for" (meaning you could tolerate and were seeking) liquidity risk, expecting of course to be compensated for the risk you were taking on, would you start your search in Canada or Singapore?

Based just on the information in the chart, you would probably look in Australia's more liquid bond market in the first scenario but in Singapore's less liquid bond market in the second scenario.

Given the top five bond markets in terms of outstanding market value discussed above, the following chart of annual issuances shouldn't come as too much of a surprise: the same top five countries dominate. Take note of the year-over-year percentage change (Y/Y % Change beneath the main part of the table): which countries' bond markets are growing at the fastest rates? A high rate of issuance growth is another broad indicator of the health of the overall credit market, though of course one still needs to evaluate bonds on an individual basis before investing. By this measure, what bond markets seem to have been struggling over this 15-year period?

Figure 3: Global Bond Issuance

Global Long-Term Bond Issuance

\$ Billions											
	Australia	Canada	China	EU27	Hong Kong	Japan	Singapore	United Kingdom	United States	Other Developed Markets	Emerging Markets
2006	185.5	266.9	179.6	3,259.5	14.2	2,417.3	12.3	927.0	5,286.0	458.8	1,305.7
2007	165.1	316.1	786.0	3,409.2	13.2	2,592.0	28.5	1,047.7	5,449.4	415.8	1,462.4
2008	175.7	382.0	524.7	3,432.3	10.7	2,430.4	21.0	1,119.1	4,501.1	384.0	872.7
2009	274.7	438.8	524.5	4,352.1	25.1	3,232.2	26.9	1,181.4	6,165.5	450.6	1,233.7
2010	320.0	472.9	704.5	3,667.3	30.2	3,594.0	37.6	674.4	6,727.2	470.4	1,753.5
2011	335.6	554.5	900.8	3,698.3	32.4	4,045.9	33.2	731.4	5,657.9	523.5	1,725.1
2012	346.7	516.9	791.5	3,727.3	45.9	4,285.9	40.9	734.7	6,067.8	541.7	1,911.9
2013	311.6	576.2	975.9	3,602.9	38.3	3,747.3	43.6	645.2	5,055.4	536.0	1,732.9
2014	314.0	547.0	1,603.3	3,668.4	53.6	3,469.1	39.5	642.5	5,272.5	560.9	1,868.5
2015	240.8	502.4	2,436.2	2,947.0	44.9	3,219.0	31.9	486.7	7,803.2	559.8	1,778.5
2016	294.0	499.8	3,650.6	2,908.3	63.0	3,425.4	51.8	489.1	6,330.0	544.7	2,197.0
2017	299.3	589.7	2,790.8	2,969.5	64.2	3,230.5	36.0	474.1	5,641.4	547.3	2,024.7
2018	258.8	524.9	2,908.9	2,797.7	46.8	3,033.4	51.9	350.4	5,502.4	545.3	1,871.7
2019	219.1	538.8	3,782.1	2,960.1	88.8	3,125.0	50.2	416.3	8,816.0	574.1	2,437.4
2020	419.7	742.5	5,088.4	3,756.3	81.9	3,364.5	37.4	752.6	9,401.9	690.7	3,000.5
Average	277.4	498.0	1,843.3	3,410.4	43.5	3,280.8	36.2	724.8	6,231.8	520.2	1,811.7
Y/Y % Change	91.6%	37.8%	34.5%	26.9%	-7.7%	7.7%	-25.5%	80.8%	9.1%	20.3%	23.1%
5-Year CAGR	11.7%	8.1%	15.9%	5.0%	12.8%	0.9%	3.2%	9.1%	3.8%	4.3%	11.0%
10-Year CAGR	2.7%	4.6%	21.9%	0.2%	10.5%	-0.7%	-0.1%	1.1%	3.4%	3.9%	5.5%

Sources: Bloomberg, Refinitiv

Note: Long-term is defined as a security with maturity of 13 months or longer at issuance. Includes corporate, municipal and sovereign issuance.

Source: Kolchin, Katie et al. 2021 *Capital Markets Fact Book*. SIFMA, July 2021, p. 37. <https://www.sifma.org/wp-content/uploads/2021/07/CM-Fact-Book-2021-SIFMA.pdf>.

At 92% and 81% year-over-year growth, the bond markets in Australia and the UK were healthy over the period in question according to this metric. By the same token, Singapore's -26% "growth" suggests some problems in that market that require further investigation.

2. Credit Spreads and Credit Default Swaps

Again, the above assessments of the overall health of credit markets are much too broad to justify an actual specific investment. A more thorough analysis might start with the credit spread. A spread is just a difference in levels; a **credit spread** is the difference in yield between a particular bond and the risk-free rate for the same maturity. Stated differently, the credit spread is the *yield premium* for a fixed maturity. A risky bond needs to yield more to attract investors. If a risk-free bond and a risky bond had the same yield, investors would only invest in risk-free bonds. Consequently, risky bonds need to yield more.

The riskier the bond, the higher the yield must be in order to attract investors. The difference in yield between a risky bond and a risk-free bond (the credit spread) is therefore a good indicator of the riskiness of the bond. In fact, the credit spread can be used to calculate a probability of default (PD)¹ as discussed in the "[CDS and CDS Spreads](#)" reading and video.

As you watch the "Credit Default Swaps" video found in the same reading, pay special attention to the calculation of PD. (You may also find the transcript of the video below the embedding to be helpful.) Remember that the "recovery rate" is how much value the creditor expects to *recover* if the bond issuer (the borrower) defaults.

We have already considered the notion of recovery rate in the previous lesson, but it does not hurt to repeat it here as a further application. Recall the 4,000 Naira loan to Nushi, collateralized by her nice watch. Let's say we estimate the watch to be worth 4,500 Naira. It would be irrational for Nushi to intentionally not repay her loan since the watch is worth more. But if she did default on (not repay) her loan, we have her watch, which we can sell to get some of our lost loan amount back. It may be the case that we aren't able to find a buyer for the watch at the 4,500 Naira price. (Watches are not that *liquid*, as we will discuss more later in this lesson.) We may only be able to sell it

quickly for 3,000 Naira, in which case we only recover 3,000 Naira out of our 4,000 Naira loan so that the recovery rate on this loan is 75%.

Once you are familiar with the PD calculation that uses credit spread (CS) and recovery rate (RR) as inputs from the video, you can calculate the probability of Nushi's default with just a few assumptions. Let's say you'll have a one-month loan term and the risk-free interest rate (RFR) is 1% per month. You decide to make Nushi the loan at the rate of 5% (so the bond yield is 5%): she should pay you back 4,200 Naira (4,000 Naira principal plus 200 Naira interest) in one month. Based on the definition of credit spread above, $CS = BY - RFR = 5\% - 1\% = 4\%$. We already calculated our recovery rate, $RR = 75\%$.

$$PD = CS / (1 - RR) = .04 / (1 - .75) = .16 = 16\%.$$

The probability of default implied by these assumptions is 16%. What if you actually believe Nushi's PD is 25%? Then, the 4% CS is not enough to compensate you for the risk. You can easily rearrange the equation above to find the credit spread that corresponds to the PD you assign to Nushi:

$$CS = PD (1 - RR)$$

Now you would require a credit spread of

$$CS = PD (1 - RR) = .25 * (1 - .75) = .0625 = 6.25\%$$

So you would require a higher interest rate of $RFR + CS = 1\% + 6.25\% = 7.25\%$, or 290 Naira.

Would the CS go up or down if you believed that Nushi's PD was lower, say, only 10%?

$$CS = PD (1 - RR) = .10 * (1 - .75) = .025 = 2.5\%, \text{ which is lower than the original } CS = 6.25\%.$$

Go ahead and manipulate the inputs in the following estimation formula and see whether the impact of the changes are linear or nonlinear.

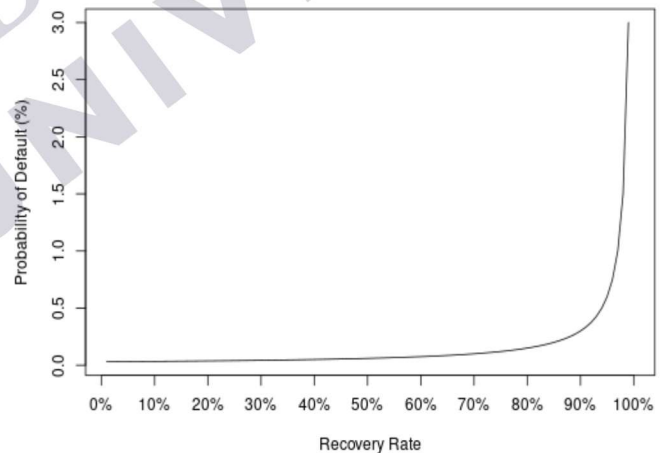
What did you find? Discuss with your classmates in the forum.

Probability of Default, Credit Risk, Credit Spread

Interest rate (%):

Risk-Free Rate (%):

Credit Spread = 3%



IMPORTANT REMARK about the PD formula. Say that $RF = 2\%$, the interest rate on the bond is 5% , so that $CS = 3\%$. Assume now that $RR = 90\%$. If we use the formula $PD = CS / (1 - RR)$, we find

$$PD = \frac{3\%}{1 - 90\%} = 0.30$$

which seems to be a very reasonable conclusion. However, if in the same setting, we change RR from 0.90 to 0.99, we find

$$PD = \frac{3\%}{1 - 99\%} = 3.$$

This clearly does not work since a probability cannot be larger than 1. So what is going on? It is very simple: the formula for PD only works locally. When we assume that CS was 3% with a RR of 90%, that could have been correct, but it is very unlikely that $CS = 3\%$ when $RR = 99\%$. In fact, one could even argue whether we need to have a credit spread/premium when, in case of default, we are able to recover nearly the entire investment.

The gist of things here is that we cannot toy with the formula completely at will because the CS and RR inputs are not independent, but correlated. Therefore, do not include $PD=3!$ which would show a complete ignorance of the laws of probability, not a good trait to exhibit for someone who wants to be a quantitative analyst.

Assume that we know the probabilities of default for x and y years, with $y > x$, and let them be PD_x and PD_y , respectively. In some applications it is important to infer the intermediate average probability of default between a y -year CDS and an x -year CDS, which we will denote by PD_{y-x} . This is not the arithmetic average of PD_x and PD_y , but the following:

$$PD_{y-x} = \frac{y \cdot PD_y - x \cdot PD_x}{y-x}.$$

Example: Suppose that the 2 year CDS spread is 80 bps and that the 5 year CDS spread is 125 bps. Assuming a recovery rate of 60% in both cases, what is the estimated average probability of default between year 2 and year 5?

Answer: We can easily calculate $PD_2 = 0.80\% / (1 - 0.6) = 2.00\%$, and $PD_5 = 1.25\% / (1 - 0.6) = 3.125\%$. Then using the formula given above we find $PD_{5-2} = \frac{5 \cdot 3.125\% - 2 \cdot 2.00\%}{5-2} = \frac{11.625\%}{3} = 3.875\%$.

As discussed in “7.1 CDS and CDS Spreads,” a CDS is very similar to traditional insurance in that the CDS buyer pays a premium to insure against the risk of a bond’s default, much like an insurance customer buys (pays a regular premium for) insurance against the risk of, for example, a catastrophe damaging a home.

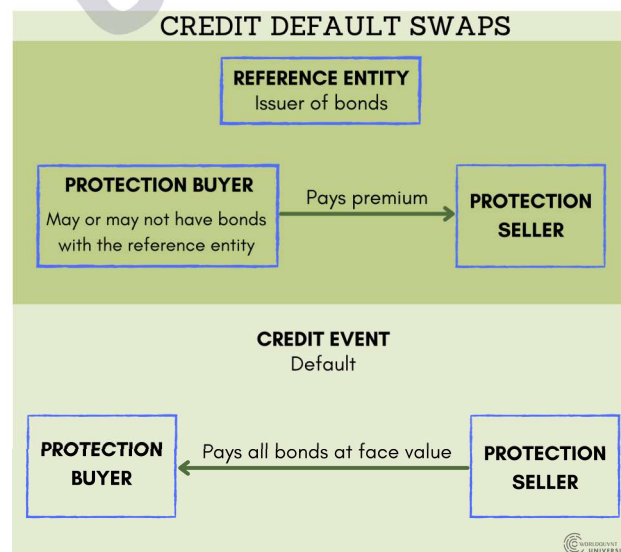
An important difference between a CDS and traditional insurance, however, is that you don’t need to own any bonds to buy a CDS. You can buy protection on something you don’t even own. In other words, you can use a CDS to *speculate* on a bond instead of just *hedging* the risk on the bond. Likewise, you can use a CDS to sell protection on something.

This is very different from the traditional insurance market. If someone else could buy fire insurance on your house, they could make a lot of money if your house burned down. They could make so much money that they might even set your house on fire. This would be an example of moral hazard (“where one party takes excessive risk for their own benefit because they will not bear the full cost of those risks”), which we discussed in the last lesson.

When you own a bond and buy CDS protection on that bond, you have the same interest rate risk from the bond that you had before, but you’ve offloaded the credit risk to some other market participant. If you don’t have the bond but you believe that the bond is headed for default, you can buy protection (on the bond that you do *not* have) and make money on the default.

This aspect of CDSs is analogous to buying a put option on an equity security (stock): you don’t need to own the stock in order to buy a put or sell a call on that stock.

Figure 4: Credit Default Swap: Protection Buyer, Protection Seller, and Reference Entity



CDSs therefore allow for the easy buying and selling of the credit risk of a specific borrower. We say that CDSs “commodify” credit risk: CDSs allow us to buy and sell credit risk as if it were a mere commodity. In other words, CDSs nicely package, price, and transfer credit risk from one party to another, turning it into a commodity.

By using [this Shiny App](#), students will have the opportunity to plot the CDS curves of several large market cap assets on several dates.

3. Ratings and Credit Spread

As we saw in the calculation examples above, higher PD implies a higher credit spread. Ratings are a function of the PD, or probability of bankruptcy as used in the table: the highest rating (AAA) is for bonds that have almost no risk of default, meaning PD at or very near zero.

Note that "investment grade" refers to bonds rated BBB or better, and bonds rated below BBB are called "junk." There is nothing disparaging about the latter term; it is just credit market jargon. Another term for these sub-BBB bonds is "high yield."

Figure 5: Ratings and Probabilities of Default

Credit rating	Probability of bankruptcy, %
AAA	0.07
AA	0.51
A+	0.6
A	0.66
A-	2.5
BBB	7.54
BB	16.63
B+	25.0
B	36.8
B-	45.0
CCC	59.01
CC	70.0
C	80.0
D	100.0

Source: Pirogova, Oksana et al. "The Algorithms for the Environmental Finance Based on Adjusted Present Value Models." *E3S Web of Conferences*, 2019, vol. 91, p. 9. https://www.e3s-conferences.org/articles/e3sconf/abs/2019/17/e3sconf_tpacce2019_08021/e3sconf_tpacce2019_08021.html.

The credit ratings on different bonds are based on the PDs that the rating agencies calculate. (Again PDs, not a single PD: rating agencies will make their own assumptions, resulting in different models and different PD results, though the various PDs are likely rather similar.) A *low* PD is associated with a *high* rating and vice versa.

Notice how the credit spread in the following table increases as the rating worsens, e.g., from BBB to BB+ or from B+ to B-. As we expect, the ratings go down as the credit spread increases because the ratings go down as the PDs increase. Notice too how the credit spread increases as you move to the right across the table, from shorter-term bonds to longer-term bonds. Investors need to be compensated for the additional risk of default due to longer-term exposure.

This table only shows credit spreads for bonds rated BBB and worse and for one to ten years. Would the spread of an A-rated one-year bond be higher or lower than 0.78%? Would the spread of a BB-rated 30-year bond be higher or lower than 3.83%?

Figure 6: Ratings, Probabilities of Default, and Time to Maturity

	1Y	3Y	5Y	10Y
BBB	0.78%	1.04%	1.24%	1.66%
BBB-	1.11%	1.47%	1.72%	2.42%
BB+	1.24%	2.08%	2.58%	3.47%
BB	1.85%	2.65%	3.07%	3.83%
BB-	1.87%	2.67%	3.09%	3.85%
B+	2.28%	3.18%	3.50%	4.14%
B	2.30%	3.37%	4.07%	4.95%
B-	2.96%	4.90%	5.81%	6.82%

Data source: Bloomberg.

Source: Montesi, Giuseppe et al. "Stochastic Optimization System for Bank Reverse Stress Testing." *Journal of Risk and Financial Management*, vol. 13, no. 174, 2020, p. 22. <https://www.mdpi.com/1911-8074/13/8/174>.

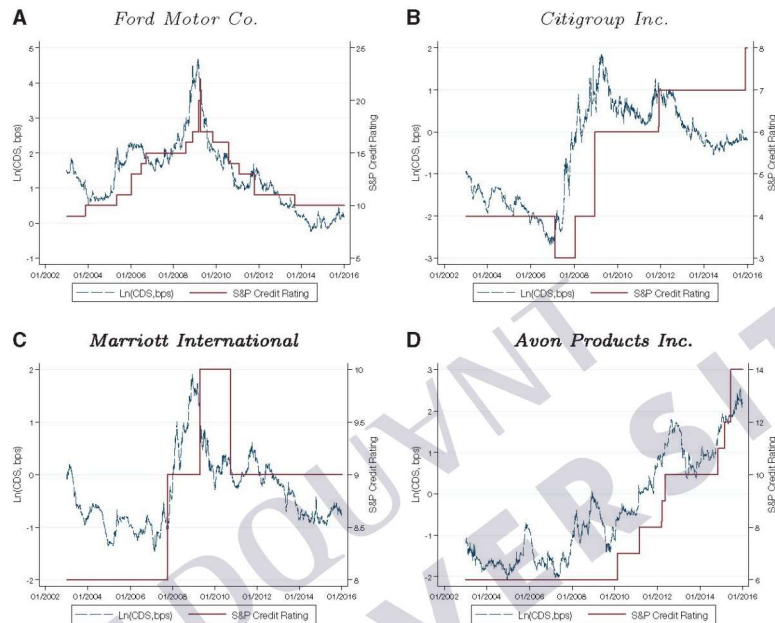
If you said "lower" for the A-rated 1-year bond and "higher" for the BB-rated 30-year bond, then you are right. If you answered anything else, spend some more time assessing the trends represented in the table. You might even notice that credit spread and time to maturity are positively correlated. Also, the credit spread is correlated with the rating: as the rating improves, the credit spread decreases. Furthermore, considering this table alongside the table above it, you might (rightly) suspect that credit spread and probability of default are positively correlated, which makes sense given the simple PD calculations we explored in the previous section of this lesson.

Let's say that the column on the right in this table is indicative of credit spreads for the ratings in the first two columns. There is a bond rated B1/B+ (B1 by Moody's and B+ by Fitch) and the current credit spread for that bond is 400. However, based on your estimation of PD, you believe the credit spread should only be 200. The market is pricing in a much higher probability of default, meaning most participants in the market are using a higher PD as

an input to their model. If you are confident in your own assumptions and inputs, you can buy this bond at a price that is lower than the value, and it will be a good investment—if you are right.

Another major problem with rating agencies is the fact that the ratings do not necessarily change in a timely manner. Ratings are almost always "lagged" (after the fact) in relation to the events or circumstances that affect the credit quality of the rated bonds. We will discuss CDS (credit default swaps) in the next lesson, but for now, consider the CDS spread to be an almost instantaneous quantification of the market's perception of credit quality. Notice in charts A through D that the CDS spread has already changed significantly before the rating change is implemented, meaning that a significant amount of time passes between when market participants recalibrate their probability of default expectations and when the rating agency actually changes its rating with an upgrade or downgrade.

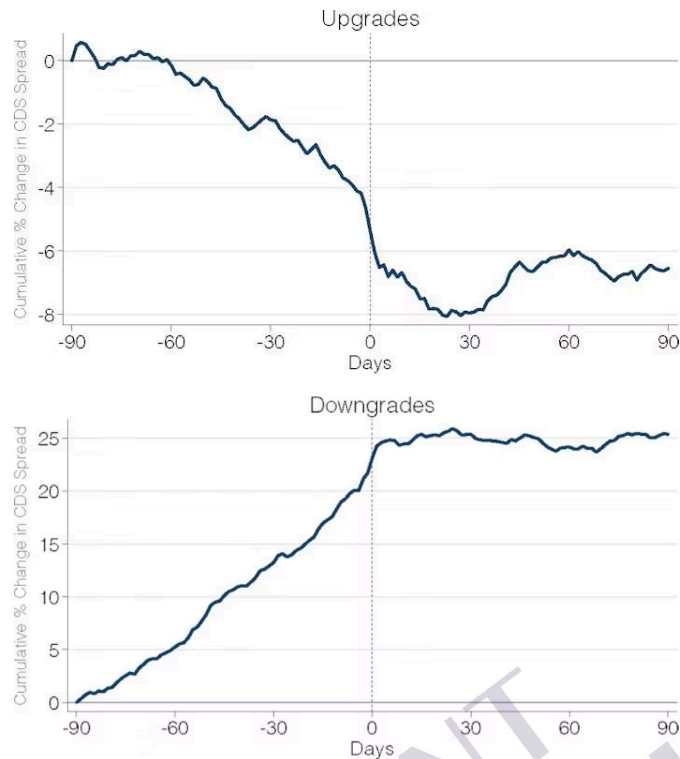
Figure 7: Credit Spread vs. Credit Rating over Time



Source: Lee, Jongsub et al. "CDS Momentum: Slow-Moving Credit Ratings and Cross-Market Spillovers." *The Review of Asset Pricing Studies*, vol. 11, no. 2, 2021, p. 354. <https://doi.org/10.1093/rapstu/raaa025>.

The following pair of charts tell much the same story, but summarizing large samples of ratings instead of individual companies, as above, changes things. In both cases, the CDS spread has almost completely stabilized at some new level before the rating agency changes take effect.

Figure 8: Credit Spread vs. Credit Rating Change



Source: Lee, Jongsub et al. "CDS Momentum: Slow-Moving Credit Ratings and Cross-Market Spillovers." *The Review of Asset Pricing Studies*, vol. 11, no. 2, June 2021, p. 355. <https://doi.org/10.1093/rapstu/raaa025>.

All of these charts attest to the **rear-view mirror** approach that rating agencies have to assessing risk: their models were reactive (reacting to changes in circumstances) instead of forward-looking and proactive. Many people object to Value-at-Risk (VaR), one of the most common and standard risk management calculations, for this reason. Indeed, this criticism aptly describes many risk calculations and techniques because they largely rely on data and statistics that summarize past events. (You will absolutely learn ways to approach the estimation of risk that are not prone to this criticism in later courses.)

Much of this problem can be described as a type of moral hazard: since the rating agency does not bear the risk of misrating, they may not be as responsive as if they had a significant investment in the bonds being rated. They do of course still bear the reputational risk of misrating. A rating agency with a bad track record of forecasting probability of default may lose revenue as their unreliability becomes better known.

The required reading "[Stakes Sensitivity and Credit Rating: A New Challenge for Regulators](#)" about the challenges inherent to the credit rating industry describes the problematic but important function of rating agencies in the investment community. They serve not only as judges and monitors of creditworthiness, but in a de facto sense, they also approve or endorse certain investments for certain types of investors. (Note: You do not need to read the abstract and the large section called "Stakes Sensitivity of Knowledge Transmission" as these delve into a more philosophical discussion of the problem that does not directly concern us; however, do read the third and final section "A Challenge for Regulators.")

4. Liquidity and Liquidity Risk

Default risk is just one of the risks that should concern us. As discussed above, bonds also have liquidity risk, which generally refers to "the risk that investors won't find a market for the bond, potentially preventing them from buying or selling when they want." We can calculate default risk (PD) in a rather straightforward way, so how do we measure or calculate liquidity risk?

First, let's define liquidity. The real value of an asset is CHF 100, but right now, you can sell it for only 90 CHF. The difference, 10 CHF here, indicates how liquid the asset is. The more liquid an asset is, the closer the market value and selling price are. If you want to sell a FAANG stock (that is, one of the big U.S. tech companies—Facebook (now Meta), Amazon, Apple, Netflix, and Google (now Alphabet)) during trading hours on the NYSE (New York Stock Exchange), you can get a price (bid price) close to the market value (average of bid and ask price). But if you want to sell an unknown stock in the after-hours market, you will get a bid price that is far from the ask price (high bid-ask spread) because it is not a highly traded stock. It is off hours, and it's very likely that there are only a few market makers around.

The authors of "Market Liquidity Risk: An Overview" (Stange and Kaserer) summarize several approaches to calculating, or modeling, liquidity risk. We will work with some of these models in the next course. For the

purposes of this lesson, you should read about the models to understand why certain factors or inputs are used the way they are.

Stange and Kaserer also provide a helpful discussion of the important aspects of liquidity. For example, liquidity risk is not merely an absent-or-present phenomenon but rather a complex and messy risk that changes over time (even with time of day) in nonlinear ways. Liquidity is relative, not binary; it isn't that a bond or other security is either liquid or not (discrete). Some bonds are more liquid than others, and that liquidity can increase or decrease over time depending on market circumstances. Also, liquidity is often a function of your position size (the amount of a security you have). Liquidity also depends on how much time you have to liquidate, meaning your liquidation horizon.

One of the inputs used in many liquidity models is the bid-ask spread, or the difference between current offer prices (to sell) and bid prices (to buy) for a particular security. When these prices match, the transaction takes place, and the new bid-ask spread is the difference between the next highest offer price and the next lowest bid price. The more liquid the security, the lower this bid-ask spread should be.

Let's test whether the bid-ask spread makes sense as a liquidity measure in some other contexts:

1. Let's say you want to sell your house. You believe it's worth \$250,000, and you're happy to sell it at that price. A potential buyer inspects your house and is willing to pay \$200,000 for it. That's a \$50,000 bid-ask spread, or 22.5% of the average between the bid and offer prices. That is a lot of money, but it is consistent with the fact that houses are known to be relatively illiquid. If you are willing to wait, then eventually, another buyer might come along and offer a higher value, closer to (or even above) the market value. If you do not have the luxury of waiting, then you may be forced to sell at the highest bid price.
2. Let's say you buy 100 shares of Apple Inc. stock at \$160 per share. In a normal market, you should be able to sell those same 100 shares later that day at very close to the price at which you bought them, say \$159 per share, only a .6% difference from when you bought it. This is consistent with the fact that the stock market for blue-chip (high-quality) stocks is generally very liquid.
3. When you tried to sell Nushi's watch after she defaulted, you would have been happy to sell it for 4,000 Naira, but the highest price you could find was 3,000 Naira, meaning the bid-ask was 1,000 Naira, or almost 30% of the average between the bid and offer prices. This confirms what we said before: watches are not that liquid, especially used watches.

In the reading "Liquidity, Credit Risk and Pricing of Corporate Bond," traded volume appears frequently in the discussion of liquidity, where lower volumes imply less liquidity. Volume is especially central to the calculation of the Amihud illiquidity factor. Using the same three examples, we can test whether volume might function as an indicator of liquidity:

1. You buy and sell your house infrequently, quite rarely in fact.
2. On the other hand, the volume of Apple shares traded every day is measured in the tens of millions.
3. You will only need to sell Nushi's nice watch once.

In all three cases, transaction volume correlates highly with what we know about the liquidity of these markets.

Another important factor in liquidity is **fungibility**—how easily a good or asset can be interchanged with another thing or asset of the same type. Let's review the three examples again with this in mind:

1. You chose your house because it has the right number of rooms and the right size; because it is located in the neighborhood you want to live in; because it has the style or architecture that you like; etc. Although it may look exactly the same as the house next door, it has different amounts of sunlight, a different location, and different views. Although there are many other houses that are the same size with the same number of rooms, even in the same neighborhood, they will not be even close to the same house. Houses are not easily interchanged for other houses. Houses are not considered fungible because no two houses can be identical. Even if everything else is almost exactly the same, they certainly have different GPS coordinates.
2. You could buy one share of Apple stock, then sell it later to your friend who already has one share of Apple stock. The next day, you can change your mind again and buy a share of Apple stock from the same friend. You wouldn't notice whether your friend sold you the share that she bought from you or the share that she has owned for longer. And you would not care because stock shares are fungible.
3. On the other hand, you would be very hesitant if someone wanted to trade (used) watches with you, even if they are the same make and model. You cannot easily tell, for example, if the other watch works as well as yours; it might take hours to ascertain whether the watch runs fast or slow. Perhaps the battery is about to run out. The watch might have some kind of damage that is not immediately apparent. Luckily, people don't often ask to trade watches because watches aren't fungible.

Note in this reading the reference to "yield spread over the benchmark curve (SP)" and the definition they provide in footnote 3. Does this spread sound like any other spread we have discussed in this lesson? It's not that similar to the bid-ask spread. In fact, this is the same as our definition of credit spread and should be interpreted as such.

5. Conclusion

In this lesson, we saw different ways of "sizing up" the credit market in terms of outstanding value and annual issuance both overall as well as by country.

We learned how to calculate a credit spread using bond yield and risk-free rate and using the PD and recovery rate. We rearranged those formulae to determine the market's view on the probability of default from a bond's yield or credit spread.

We learned how a CDS effectively isolates and commodifies credit risk. We developed familiarity with the relationships between rating and probability of default and between rating and credit spread. We discussed how credit ratings (and other risk calculations, such as VaR) are backward-looking if they rely entirely on past information.

Finally, we discussed the problem of liquidity in more detail and learned a few approaches to estimating it based on market data and some other assumptions.

In the next lesson, we apply the concept of correlation to certain aspects of the markets we have discussed: the stock and real estate markets. We also explore the nature of this correlated relationship, which leads us to a discussion of leverage and its risks. This fresh set of leverage risks leads us back to a discussion of the need for regulation.

Footnotes

1. Note that using the credit spread is just one approach to calculating the PD, hence why there is *a* probability of default and not *the* probability of default. We will discuss in more detail how different models may be more or less appropriate depending on the information available, the assumptions we consider valid, and other factors. For now, suffice it to say that the risk of not using the most appropriate model is called "model risk," which we will also discuss at much greater length in later courses.

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